

Reviewer Comments

Predicting Effective Heuristics and Optimality Gap for the Traveling Salesman Problem Using Lightweight Instance Features

Summary: This paper proposes a new approach to the Traveling Salesman Problem (TSP) by introducing a two-stage machine learning framework that predicts both the most suitable heuristic and the expected optimality gap using low-cost structural instance features. The authors construct a dataset comprising 160 TSP instances, including Euclidean, GEO, and synthetically generated Explicit instances, and evaluate four classical heuristics: Nearest Neighbor, Greedy, Christofides, and Insertion. Three machine learning models, namely Logistic Regression, Random Forest, and XGBoost, are employed for heuristic selection, while Quantile Regression and Conformal Prediction are utilized for uncertainty estimation in optimality-gap prediction. Among the evaluated pipelines, the Random Forest model achieved the best performance, reporting approximately 80% heuristic-selection accuracy, 87.5% prediction accuracy, an integrated MAE of 3.99 percentage points, and 90.62% prediction interval coverage. The authors further investigate feature importance using both Gini importance and permutation importance, identifying triangle inequality statistics and normalized MST weight as the most influential predictors. Overall, the manuscript addresses an interesting and relevant problem in combinatorial optimization by integrating heuristic selection with uncertainty estimation. However, the study is limited by a relatively small benchmark dataset, restricted heuristic diversity, limited robustness validation, and insufficient experimental analyses, all of which should be addressed before the manuscript can be considered for publication.

Major Issues – Must Fix:

1. Please remove section numbers, bullet points, and numbered points from the manuscript.
2. The entire study is developed using only 160 TSP instances, of which 128 are used for training and only 32 instances for final testing. Furthermore, the dataset is artificially expanded by generating 56 synthetic Explicit instances specifically to reduce Christofides dominance. Since these generated instances do not necessarily represent realistic TSP distributions, the reported performance (80% heuristic-selection accuracy and 3.99 percentage-point MAE) may not generalize to real-world TSP instances. A substantially larger and more diverse benchmark should be used to validate the proposed framework. If this is not feasible, the authors should explicitly acknowledge this limitation and temper the generalizability of their conclusions accordingly.
3. The manuscript evaluates the proposed framework using only four classical construction heuristics (Nearest Neighbor, Greedy, Christofides, and Cheapest Insertion). However, many state-of-the-art TSP heuristics and metaheuristics, such as Lin-Kernighan-Helsgaun (LKH), 2-opt, 3-opt, Guided Local Search, Simulated Annealing, Genetic Algorithms, and Ant Colony Optimization, are not considered. Consequently, the claim that the framework predicts the “best heuristic” should be limited to the evaluated heuristic pool and should not be generalized beyond the algorithms included in this study.
4. It was found that the proposed heuristic selector has a cross-validation accuracy of approximately 80%. Nevertheless, the final results are generated by a single train-test split of 80 to 20 using Seed 1 for the entire two-stage process, which is why the robustness of the proposed pipeline cannot be reliably assessed. As the authors give results on optimality gap prediction (MAE of 3.99 percentage points and R^2 of 0.9407) obtained from one random partition, it is recommended to conduct repeated hold-out validation or nested cross-validation to guarantee the stability of the results received using distinct data parts. The single split is also reused to choose the winning Stage 2 pipeline. The manuscript compares RF+Quantile, RF+Conformal, XGB+Quantile, and XGB+Conformal on the same held-out test set and then selects RF+Quantile because it is best on every Stage 2 metric, which means those 32 instances are functioning as model-selection data rather than an untouched test set.

This is compounded by inconsistent Stage 1 reporting. Section 3.1.2 gives Random Forest cross-validation accuracy as $80.00\% \pm 6.43\%$ on the 128 training instances, while the Seed-1 analysis elsewhere reports $76.68\% \pm 8.92\%$ for the same quantity. Please provide an explicit diagram or pseudocode showing exactly which instances are used for feature selection, hyperparameter tuning, classifier fitting, gap-regressor fitting, conformal calibration, and final testing, and restructure the evaluation with grouped nested cross-validation, or a single final test set that stays untouched until every model and hyperparameter choice is frozen, so that the headline MAE, R^2 , and coverage figures come from that untouched evaluation rather than from the split used to decide which pipeline wins.

5. The manuscript proposes a two-stage framework in which the heuristic prediction produced in Stage 1 is applied to Stage 2 as a new input feature for the future prediction. Still, it remains unclear how errors appearing in Stage 1 affect the result of the optimality gap prediction in Stage 2 because it is not mentioned in the manuscript. Therefore, an appropriate ablation analysis must be performed to compare the Stage 2 performance using ground-truth heuristic labels and predicted heuristic labels based on actual heuristics and those entered using the predictions made in Stage 1 and see how important each of the stages is.
6. Several reported numbers are internally impossible or inconsistently described, and the raw prediction tables are needed to audit the results. An exact-match count over 160 instances can only move in steps of $1/160$, or 0.625% , so the stated 80.50% exact match rate across all 160 instances cannot be an exact classifier accuracy on that denominator, even though the displayed XGBoost folds of 0.8750, 0.8125, 0.7188, 0.8125, and 0.7812 do average to 80.00% with a population standard deviation of about 5.08% . Please also clarify whether the reported mean regret of 0.0278 is averaged over all instances, in which case correct predictions contribute zero and the conditional regret among mistakes must be larger than the 2.78% quoted as the loss when wrong, or averaged only among mistakes, in which case the text should say so. Please correct the Random Forest section's use of CV to mean coefficient of variation where it is being used to mean cross-validation, and reconcile the XGBoost section's training accuracy of $96.88\% \pm 0.70\%$ with Table 3's single train-accuracy value. Please provide the 160-row Stage 1 out-of-fold prediction file and the 32-row final Stage 2 prediction file, including instance, true and predicted best heuristic, all four costs and runtimes, regret, true gap, point prediction, and interval bounds, so these discrepancies can be checked directly.
7. It is claimed in manuscript that lightweight structural features can guide selection of heuristic without running costly heuristics, but calculations related to extraction of several offered features are not evaluated properly. Triangle inequality violation rate and average violation magnitude require usage of Monte Carlo sampling on 200,000 of random triples, and maximum spanning tree computations add to its cost. The manuscript does not compare the preprocessing cost of feature extraction with the computational time saved by avoiding heuristic evaluation. Thus, the claimed computational efficiency of the suggested framework is not sufficiently verified. The asymptotic costs make this more than an unmeasured claim. Several of the ten features carry the same complexity class as the heuristics they are meant to replace: the edge-statistic and minimum-spanning-tree features are $O(n^2)$, matching Nearest Neighbor, and `pct_short_edges` is $O(n^2 \log n)$, matching Greedy, while the MST feature computes a central subproblem of Christofides itself. Please report per-instance wall-clock time, on the same hardware, for the complete feature vector, including the 200,000-triple Monte Carlo calculation, each of the four heuristics, and Stage 1 and Stage 2 inference, and state the actual end-to-end speedup this yields, since equal or worse asymptotic cost would mean the framework saves no computation even at high selection accuracy.
8. Stage 2 treats four heuristic rows per TSP instance as independent observations, so 640 rows correspond to only 160 independent feature vectors, and the conformal calibration procedure may mix rows from the same instance across model fitting and calibration. The manuscript states that 20% of the training data is withheld as a calibration set, 100 rows, leaving 398 rows

for the regressor, which reads as a row-level split. If so, one heuristic row for a TSP could train the regressor while another row from the identical instance, carrying the same ten structural features, sits in calibration, which is not the independent calibration experiment the conformal argument assumes. Please state whether the calibration split, and any other resampling, tuning, or interval calculation in Stage 2, was made by TSP instance or by row, report it in numbers of instances rather than rows, and rerun the analysis with groups defined by instance if the split was row-wise. The effective independent sample size is at most 128 training TSPs for the Seed-1 split, not 398 or 498 independent observations, and the reported figures should reflect that.

9. Several input features are not invariant to a trivial rescaling of the same TSP, which the prediction target should not depend on. Multiplying every edge weight by a positive constant does not change which heuristic produces the best tour and does not change percentage optimality gaps, yet `min_edge_weight`, `std_edge_weight`, and `avg_violation_magnitude` are expressed in raw distance units, with the latter reaching 4,629 in the manuscript's own `brg180` example. `StandardScaler` does not fix this, since it standardizes a column across the dataset rather than normalizing each instance against its own arbitrary choice of distance units, so tree models can learn thresholds tied to a particular dataset's unit conventions rather than genuine geometric structure. Please rescale every instance by 0.01, 10, and 1000, recompute the features, and confirm heuristic and gap predictions are unchanged. If they are not, normalize the dimensionful features by an instance-level scale such as mean or median edge weight and repeat the full analysis. Please also give the exact formula for `pct_short_edges`, since if its threshold is computed as the same percentile within every instance, the fraction below it is close to fixed by construction and would not be a meaningful feature.
10. The manuscript concludes that Random Forest outperforms the competing models; however, no statistical significance testing is performed to determine whether the observed improvements in heuristic-selection accuracy or optimality-gap prediction are statistically meaningful. Appropriate statistical tests should be included before claiming the superiority of the proposed framework.
11. This paper highlights how explicit TSP cases mitigate Christofides's advantage through the creation of 56 synthetic instances. Nevertheless, the work does not check how the number of synthetic instances impacts classifiers' efficiency. All in all, testing the model with varying ratios of synthetic to actual instances would increase the reliability of the findings by ensuring that the deduced enhancements cannot only be explained by the dataset. The risk goes beyond ratio sensitivity to what the classifier is actually learning. Instance type is itself an input feature, the type-only rule already reaches 68.13% accuracy on its own, and `type_Explicit` is the most important XGBoost Gini feature at 0.200, so part of the reported 80% may reflect the model recognizing which generator produced an instance rather than a transferable structure-to-heuristic relationship. This is especially likely if synthetic instances that differ only in size or random seed end up split across training and testing. Please add grouped cross-validation that holds complete synthetic families or generators out together, ideally leave-one-family-out, a TSPLIB-only evaluation in which no generated instance appears in the test distribution, and an ablation that removes the type feature to show how much predictive power remains without it.
12. Please expand the literature review and include at least 25–30 recent peer-reviewed primary research articles to strengthen the academic foundation of the manuscript.
13. Several citations are misrepresented, inaccurate, or over-extended to support claims their sources do not establish. Please audit every citation in the manuscript against the specific claim it is attached to, not merely the general topic.

Major Issues – Recommended:

1. Even though the research paper considers heuristic selection performance and optimality-gap estimation independently, it fails to give any specific error assessment of the cases where the heuristic selection model failed. A study of those instances, which includes description of the characteristics of the incorrectly classified TSP instances and reasons of incorrect heuristic selection, would bring useful information about the limitations of the proposed approach.
2. The research paper provides a general heuristic-selection accuracy of about 80%, but it lacks a class-wise analysis. Since the dataset includes four classes of heuristics that differ in their representation, it would be better to present precision, recall, F1-score for each class, and the confusion matrix in order to get insights on the areas of classifier success and failure.
3. The research includes various graph-structural features, such as convex hull ratio, MST weight, clustering coefficient, and triangle inequality violation data. However, no correlation or multicollinearity analysis is performed. As highly correlated features can provide the same data, addition of correlation matrix or variance inflation analysis would enhance the understanding of selected features.
4. The task of investigating only instances of the Euclidean Traveling Salesman Problem has been taken. This may not be a problem in light of the performed experiments, but discussion should be offered on the subject of whether the proposed feature-centric heuristic selection model can be applied to asymmetric, non-Euclidean, or real-life routing problems. A description of expected limitations of such use should also be presented.
5. The focus of the comparison is mainly made on the prediction effectiveness (accuracy, MAE, RMSE), but there is no practical review pertaining to the resulting tours of the suggested approach in line with the use of Christofides' method for all samples. It would be useful to include such comparison to allow for determining the level of advantage given by adaptive heuristic selection.
6. The manuscript delivers Random Forest feature importance and Quantile Regression metrics, but it does not analyze the sensitivity of the proposed models to important hyperparameters. A hyperparameter sensitivity analysis would demonstrate that the reported performance is not dependent on a single hyperparameter configuration.
7. The framework indicates both the most efficient heuristic and the expected optimal gap, however, the authors do not give any applied guideline demonstrating how to put this information into practice. The inclusion of a simple inference pipeline or decision procedure would increase the usability of the methodology.

Minor Issues:

1. The paper mentions that out of the 640 heuristics that were implemented, 23 were disregarded due to the run time being longer than the limit of 600,000 ms imposed. Nonetheless, the manuscript does not specify the specific TSP case and heuristic.
2. The computations presented in Table 2 correspond to the instance feature complexity and generate the Monte Carlo estimates based on 200,000 random triples to calculate the triangle inequality. It should be noted that $O(S)$ notation has been introduced, however, the letter S is never defined.
3. The paper provides a number of benchmark results such as that of the Single Best Solver (46.88%) and the Type-Based Rule (68.13%). Including these benchmark results in the same table as the proposed models would make the performance improvements easier to interpret.
4. In Figures 2, 3, and later ones the authors calculate Gini and permutation importances. However, it would have been nice to provide the actual numbers for feature importance as well as their display in a supplementary table.
5. The author uses several examples such as brg180, gr21, eil76, fri26, si535, and christofides hard n50 s0 in order to illustrate the behavior of the proposed framework. In order to enhance the effectiveness of the discussion, the author should provide a separate table

containing successful and unsuccessful cases with their predicted heuristic, actual heuristic, and regret values.

6. Stage 2 evaluation shows both median (14.98pp) and mean (358.19pp) widths for Quantile Regression. In order to visualize distribution of the interval lengths, it would be beneficial to include a histogram or boxplot of the prediction intervals.
7. Some empirical coverage values for Quantile Regression (90.62%) and Conformal Prediction (75.00% and 78.12%) are presented in the manuscript. For the sake of better interpretation, plotting calibration plots comparing the nominal and empirical coverage would be useful.
8. Several abbreviations, including CV, MAE, RMSE, QR, R^2 , MST, and CH, appear throughout the manuscript. Although most are eventually explained, providing a table of abbreviations would improve readability, particularly for readers from the optimization community who may be less familiar with the uncertainty quantification terminology.
9. Please ensure that all figures and tables are properly labeled, sequentially numbered, referenced within the text, and accompanied by clear and descriptive captions/titles.

Manuscript Formatting Requirements

- **Font:** Times New Roman, 12 pt (headings should be bold).
- **Manuscript Length:** Maximum 20 pages with single line spacing, or 40 pages with double line spacing.